Vector Fields and Line Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector fields and divergence**

For the vector field $\mathbf{F}(x, y) = \langle x^2 y,\ xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(xy^2) = 2xy + 2xy = 4xy.$

2 **[Foundational]** • **Target: Vector fields and divergence**

For $\mathbf{F}(x, y, z) = \langle y,\ -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.

$\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(-x),\ \frac{\partial}{\partial z}(y) - \frac{\partial}{\partial x}(z^2),\ \frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y) \right\rangle = \langle 0, 0,\ -2 \rangle.$

3 **[Foundational]** • **Target: Line integrals**

Evaluate the line integral $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

Parametrize $x = 3t$, $y = 4t$, $0 \le t \le 1$: $ds = 5\, dt$. $\int_0^1 (3t + 4t)(5)\, dt = 35 \int_0^1 t\, dt = \frac{35}{2}$.

4 **[Foundational]** • **Target: Line integrals**

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \le t \le 1$.

$\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t \rangle$, $\mathbf{r}'(t) = \langle 1, 2t \rangle$: $\int_0^1 (t^2 + 2t^2)\, dt = \int_0^1 3t^2\, dt = 1.$

5 **[Foundational]** • **Target: Conservative vector fields and potential functions**

Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

$P = 2xy$: $\frac{\partial P}{\partial y} = 2x$. $Q = x^2 + 3y^2$: $\frac{\partial Q}{\partial x} = 2x$. Since these are equal, $\mathbf{F}$ is conservative.

6 **[Foundational]** • **Target: Conservative vector fields and potential functions**

Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.

$f_x = 2xy \Rightarrow f = x^2 y + g(y)$. Then $f_y = x^2 + g'(y) = x^2 + 3y^2$, so $g'(y) = 3y^2$, giving $g(y) = y^3$. Potential function: $f(x, y) = x^2 y + y^3$ (plus any constant).

7 **[Foundational]** • **Target: The Fundamental Theorem for line integrals**

Using the potential function $f(x, y) = x^2 y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.

$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = (1 \cdot 2 + 8) - (0) = 10.$

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Green's Theorem**

Use Green's Theorem, $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, to evaluate $\oint_C xy\,dx + x^2\,dy$ where C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$.

$P = xy$, $Q = x^2$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - x = x$. $\iint_D x\,dA = \int_0^1 \int_0^1 x\,dy\,dx = \int_0^1 x\,dx = \frac{1}{2}$.

9 **[Intermediate]** • **Target: Green's Theorem**

Use Green's Theorem to find the area of the region enclosed by a curve C, using the formula $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$.

$dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$.
$x\,dy - y\,dx = 2\cos t(2\cos t\,dt) - 2\sin t(-2\sin t\,dt) = 4\cos^2 t\,dt + 4\sin^2 t\,dt = 4\,dt$.
$A = \frac{1}{2}\int_0^{2\pi} 4\,dt = 4\pi$ (matching the known area of a radius-2 circle).

10 **[Intermediate]** • **Target: Green's Theorem**

Use Green's Theorem to evaluate $\oint_C (3y)\,dx + (5x)\,dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.

$P = 3y$, $Q = 5x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 5 - 3 = 2$. $\iint_D 2\,dA = 2 \cdot 4\pi = 8\pi$, using that the disk of radius 2 has area 4π.

11 **[Intermediate]** • **Target: Work done by a force field**

A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.

Leg 1 ($y = 0$, $x : 0 \to 2$): $\mathbf{F} = \langle 0, x \rangle$, $d\mathbf{r} = \langle dx, 0 \rangle$: contributes $\int_0^2 0\,dx = 0$. Leg 2 ($x = 2$, $y : 0 \to 3$): $\mathbf{F} = \langle y, 2 \rangle$, $d\mathbf{r} = \langle 0, dy \rangle$: contributes $\int_0^3 2\,dy = 6$. Total work $= 0 + 6 = 6$.

12 **[Intermediate]** • **Target: Work done by a force field**

Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

$W = f(2, 3) - f(0, 0) = 6 - 0 = 6$, matching the piecewise calculation — confirming $\mathbf{F}$ is indeed conservative with potential $f(x, y) = xy$.

13 **[Intermediate]** • **Target: Testing for conservative fields**

Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.

$P = y^2$: $\frac{\partial P}{\partial y} = 2y$. $Q = 2xy + 1$: $\frac{\partial Q}{\partial x} = 2y$. Equal, so conservative. $f_x = y^2 \Rightarrow f = xy^2 + g(y)$;
$f_y = 2xy + g'(y) = 2xy + 1$, so $g'(y) = 1$, $g(y) = y$. Potential function: $f(x, y) = xy^2 + y$.

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14 [Intermediate] • Target: Testing for conservative fields

Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

$P = 3x^2y$: $\frac{\partial P}{\partial y} = 3x^2$. $Q = x^3 + 2y$: $\frac{\partial Q}{\partial x} = 3x^2$. Equal, so conservative. $f_x = 3x^2y \Rightarrow f = x^3y + g(y)$; $f_y = x^3 + g'(y) = x^3 + 2y$, so $g'(y) = 2y$, $g(y) = y^2$. Potential function: $f(x, y) = x^3y + y^2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 9.01 Separable Differential Equations

Use Green's Theorem to evaluate the circulation $\oint_C (-y)\,dx + x\,dy$ around the boundary of the rectangle $0 \le x \le 3$, $0 \le y \le 2$, traversed counterclockwise.

$P = -y$, $Q = x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2$. $\iint_D 2\,dA = 2(3)(2) = 12$.

16 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 1.01 Precalculus Review for Calculus

Evaluate $\int_C y\,dx + x\,dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.

Leg 1 ($y = 0$, $x : 0 \to 1$): $y\,dx + x\,dy = 0\,dx + x(0) = 0$: contributes 0. Leg 2 ($x = 1$, $y : 0 \to 1$): $y(0) + 1\,dy$: contributes $\int_0^1 dy = 1$. Total $= 0 + 1 = 1$.

17 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.

$f(1, 1) - f(0, 0) = 1 - 0 = 1$, matching the piecewise calculation.

18 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 11.01 Sequences and Series

Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n}\,ds = \iint_D \nabla \cdot \mathbf{F}\,dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \le 4$.

$\nabla \cdot \mathbf{F} = 1 + 1 = 2$. $\iint_D 2\,dA = 2 \cdot (\pi \cdot 2^2) = 8\pi$.

19 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.01 Convergence Tests

A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.

Test: $P = 2xy^3$: $\frac{\partial P}{\partial y} = 6xy^2$. $Q = 3x^2y^2 + 4$: $\frac{\partial Q}{\partial x} = 6xy^2$. Equal, so conservative. Potential: $f_x = 2xy^3 \Rightarrow f = x^2y^3 + g(y)$; $f_y = 3x^2y^2 + g'(y) = 3x^2y^2 + 4$, so $g'(y) = 4$, $g(y) = 4y$: $f(x, y) = x^2y^3 + 4y$. Work $= f(2, 1) - f(0, 0) = (4 + 4) - 0 = 8$.

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20 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.02 Power Series and Taylor Series

For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:

a Find the divergence $\nabla \cdot \mathbf{F}$. $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy + 2z) = 0 + 0 + 2 = 2$.

b Find the curl $\nabla \times \mathbf{F}$.

$\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(xy + 2z) - \frac{\partial}{\partial z}(xz),\ \frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy + 2z),\ \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz) \right\rangle = \langle x - x,\ y - y,\ z - z \rangle = \langle 0, 0, 0 \rangle$.

c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D. Since $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, $\mathbf{F}$ is conservative (irrotational) in 3D.

(a) $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy + 2z) = 0 + 0 + 2 = 2$.; (b) $\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(xy + 2z) - \frac{\partial}{\partial z}(xz),\ \frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy + 2z),\ \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz) \right\rangle = \langle x - x,\ y - y,\ z - z \rangle = \langle 0, 0, 0 \rangle$.; (c) Since $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, $\mathbf{F}$ is.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector Fields, Divergence, and Curl

Compute the divergence $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = \langle x^2 y, 3yz, -2xz \rangle$.

a. $2xy + 3z - 2x$ ✓

b. $2xy + 3y - 2z$

c. $x^2 + 3z - 2x$

d. 0

$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(3yz) + \frac{\partial}{\partial z}(-2xz) = 2xy + 3z - 2x$.

22 [Foundational] • Target: Vector Fields, Divergence, and Curl

Compute the 2D curl of $\mathbf{F} = \langle -y, x \rangle$.

a. 2 ✓

b. 0

c. 1

d. -2

$\text{curl}(\mathbf{F}) = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) = 1 - (-1) = 2$.

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23 [Foundational] • Target: Vector Fields, Divergence, and Curl

If a vector field has $\nabla \times \mathbf{F} = \mathbf{0}$ throughout a simply connected domain, what is it called?

a. Conservative (or irrotational) ✓

b. Incompressible

c. Solenoidal

d. Harmonic

A curl-free vector field on a simply connected domain is conservative, meaning $\mathbf{F} = \nabla f$ for some scalar potential function f.

24 [Foundational] • Target: Line Integrals with Respect to Arc Length

State the formula for $\int_C f(x, y)\, ds$ along a curve parameterized by $\mathbf{r}(t)$ for $a \le t \le b$.

a. $\int_a^b f(x(t), y(t))\sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$ ✓

b. $\int_a^b f(x(t), y(t))\, dt$

c. $\int_a^b f(x(t), y(t))[x'(t) + y'(t)]\, dt$

d. $\int_a^b f'(t)\, dt$

Since $ds = ||\mathbf{r}'(t)||dt = \sqrt{(x')^2 + (y')^2}\, dt$, the scalar line integral is $\int_a^b f(\mathbf{r}(t))||\mathbf{r}'(t)||dt$.

25 [Foundational] • Target: Line Integrals with Respect to Arc Length

Evaluate $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

a. $\frac{35}{2}$ ✓

b. 35

c. 7

d. 12

Parametrize: $x = 3t, y = 4t$ for $t \in [0, 1]$. $ds = \sqrt{9 + 16}\, dt = 5\, dt$.
$\int_0^1 (3t + 4t)(5)\, dt = 35\int_0^1 t\, dt = 35[t^2/2]_0^1 = 35/2$.

26 [Foundational] • Target: Line Integrals with Respect to Arc Length

Find the mass of a wire bent in the shape of the semicircle $\mathbf{r}(t) = \langle 2\cos t, 2\sin t\rangle$ $(0 \le t \le \pi)$ with density $\rho(x, y) = y$.

a. 4 ✓

b. 8

c. 2π

d. 4π

$ds = 2\, dt$. $M = \int_0^\pi (2\sin t)(2)\, dt = 4[-\cos t]_0^\pi = 4[-(-1) - (-1)] = 4(2) = 8$.

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27 [Foundational] • Target: Line Integrals of Vector Fields (Work Integrals)

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, -x \rangle$ along the circle $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$ for $0 \le t \le 2\pi$.

a. -2π ✓

b. 2π

c. 0

d. π

$\mathbf{F}(\mathbf{r}(t)) = \langle \sin t, -\cos t \rangle$. $d\mathbf{r} = \langle -\sin t, \cos t \rangle dt$. Dot product: $-\sin^2 t - \cos^2 t = -1$. $\int_0^{2\pi} (-1)\, dt = -2\pi$.

28 [Foundational] • Target: Line Integrals of Vector Fields (Work Integrals)

Find the work done by the constant force $\mathbf{F} = \langle 3, 5 \rangle$ moving an object along the straight line from $(1, 1)$ to $(4, 5)$.

a. 29 ✓

b. 24

c. 32

d. 15

Displacement $\mathbf{d} = \langle 4 - 1, 5 - 1 \rangle = \langle 3, 4 \rangle$. Work $= \mathbf{F} \cdot \mathbf{d} = (3)(3) + (5)(4) = 9 + 20 = 29$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Line Integrals of Vector Fields (Work Integrals)

Evaluate $\int_C (x\, dx + y\, dy)$ along the parabolic arc $y = x^2$ from $(0, 0)$ to $(1, 1)$.

a. 1 ✓

b. 1/2

c. 2/3

d. 0

Notice $\mathbf{F} = \langle x, y \rangle = \nabla(\frac{1}{2}x^2 + \frac{1}{2}y^2)$. By FTLI: $f(1, 1) - f(0, 0) = (1/2 + 1/2) - 0 = 1$.

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30 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

State the Fundamental Theorem of Line Integrals for a conservative field $\mathbf{F} = \nabla f$.

a. $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ ✓

b. $\int_C \nabla f \cdot d\mathbf{r} = 0$ for all curves

c. $\int_C \nabla f \cdot d\mathbf{r} = ||\nabla f||$

d. $\int_C \nabla f \cdot d\mathbf{r} = f'(\mathbf{r}(b))$

The work done by a conservative vector field depends only on the endpoints: $\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$.

31 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

State Green's Theorem relating a line integral along a positively oriented boundary ∂D to a double integral over D.

a. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$ ✓

b. $\oint_C (P\,dx + Q\,dy) = \iint_D (P + Q)\,dA$

c. $\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$

d. $\oint_C (P\,dx + Q\,dy) = 0$

Green's Theorem connects circulation around a closed curve to the flux of the curl across the enclosed region: $\oint_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA$.

32 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the unit circle $x^2 + y^2 = 1$ oriented counterclockwise.

a. 2π ✓

b. 8π

c. 0

d. π

$P = 3y, Q = 5x \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 5 - 3 = 2$. By Green's Theorem: $\iint_D 2\,dA = 2 \times \text{Area}(D) = 2(\pi \cdot 1^2) = 2\pi$.

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33 [Intermediate] • Target: Line Integrals with Respect to Arc Length

Find the work done by $\mathbf{F} = (x^2, y^2, z^2)$ along line segment from $(0, 0, 0)$ to $(1, 1, 1)$:

a. 1 ✓

b. $\frac{2}{3}$

c. 3

d. $\frac{1}{2}$

Parameterizing the segment by $\mathbf{r}(t) = (t, t, t)$ for $0 \le t \le 1$, the work is $W = \int_0^1 (t^2 + t^2 + t^2)\, dt = 3\left[\frac{t^3}{3}\right]_0^1 = 1$.

34 [Intermediate] • Target: Line Integrals of Vector Fields (Work Integrals)

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$ ✓

c. 4π

d. 12π

Applying Vectors and Vector-Valued Functions principles, $3 \cdot V = 3 \cdot 4\pi/3 = 4\pi$. Therefore, the verified correct option is $3V = 4\pi$.

35 [Intermediate] • Target: Fundamental Theorem of Line Integrals and Green's Theorem

Work done by $\mathbf{F} = (y, x, 0)$ along line from $(0, 0, 0)$ to $(1, 1, 0)$:

a. 0

b. 2

c. 1 ✓

d. $1/2$

$\mathbf{F} = abla(xy)$ conservative; $W = xy|_{(0,0)}^{(1,1)} = 1$.

36 [Intermediate] • Target: Vector fields and divergence

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2 + y^2 + z^2) dV =$

a. $4\pi/3$

b. 3π

c. $12\pi/5$ ✓

d. π

$\int_0^1 3r^2 \cdot 4\pi r^2 dr = 12\pi \int_0^1 r^4 dr = 12\pi/5$.

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37 [Intermediate] • Target: Line integrals

Green's theorem in the form $\iint_D (\partial Q/\partial x - \partial P/\partial y)dA = \oint_C P dx + Q dy$. For $P = 0, Q = x$: $\oint_C x\,dy =$

a. $-\iint_D dA$

b. 0

c. $\iint_D dA = \text{Area}(D)$ ✓

d. $\oint_C y\,dx$

$Q_x = 1 \Rightarrow \iint 1\,dA = \text{Area}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conservative vector fields and potential functions • Cross-Lesson: 11.01 Sequences and Series

Green's theorem: $\oint_C (P\,dx + Q\,dy) = \iint_D (Q_x - P_y)\,dA$. For $P = -y, Q = x$:

a. **area** $= \frac{1}{2}\oint (x\,dy - y\,dx)$ ✓

b. area $= \iint dA = 2$ times the line integral

c. perimeter $= \oint ds$

d. zero

Applying Vectors and Vector-Valued Functions principles, $\oint (xdy - ydx)/2 =$ area of D. Therefore, the verified correct option is area $= \frac{1}{2}\oint (x\,dy - y\,dx)$.

39 [Challenging] • Target: The Fundamental Theorem for line integrals • Cross-Lesson: 12.01 Convergence Tests

A conservative vector field $\mathbf{F} = ablaf$ has $abla \times \mathbf{F} =$

a. $\mathbf{F}$

b. f

c. $ablaf$

d. $\mathbf{0}$ ✓

Applying Vectors and Vector-Valued Functions principles, Conservative ■ curl-free. Therefore, the verified correct option is $\mathbf{0}$.

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40 [Challenging] • Target: Green's Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series

Line integral of scalar field: $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\,ds =$

a. area

b. volume

c. 0

d. **arc length ✓**

Applying Vectors and Vector-Valued Functions principles, $\int_C ds = L$ (arc length). Therefore, the verified correct option is arc length.

41 [Challenging] • Target: Work done by a force field • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Normal line to $z = x^2 + y^2$ at $(1, 1, 2)$: direction:

a. $(2, 2, -1)$ ✓

b. $(1, 1, 1)$

c. $(-2, -2, 1)$

d. $(1, 0, -2)$

Normal to surface $F = z - x^2 - y^2 = 0$: $\nabla F = (-2x, -2y, 1) = (-2, -2, 1)$. Direction: $(-2, -2, 1)$ or $(2, 2, -1)$.

42 [Challenging] • Target: Testing for conservative fields • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Find the flux of $\mathbf{F} = \langle x, y, z\rangle$ through the closed unit sphere using the Divergence Theorem.

a. 2π

b. $\frac{4\pi}{3}$

c. 4π ✓

d. $\frac{8\pi}{3}$

$\nabla \cdot \mathbf{F} = 3$. Volume of unit sphere $= \frac{4\pi}{3}$. Flux $= 3 \cdot \frac{4\pi}{3} = 4\pi$.

43 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 15.01 Double Integrals

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2\rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 2$.

a. 32π

b. 8π

c. -16π

d. 16π ✓

$\nabla \cdot \mathbf{F} = 2x + 2y + 2z$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\,dV = 16\pi$.

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44 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 2.01 Evaluating Limits

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle xz, 0, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \le 1,\ 0 \le z \le 1$.

a. 3π

b. $\frac{3\pi}{4}$

c. $\frac{3\pi}{2}$ ✓

d. $-\frac{3\pi}{2}$

$abla \cdot \mathbf{F} = 3z$. The flux equals $\iiint_E abla \cdot \mathbf{F}\, dV = \frac{3\pi}{2}$.

45 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 2.02 Continuity

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 3$.

a. 36π ✓

b. 72π

c. 18π

d. -36π

$abla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E abla \cdot \mathbf{F}\, dV = 36\pi$.

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